

INTRODUCTION

I received my **M.S. from Pohang University of Science and Technology (POSTECH)** in August 2025. Since September 2025, I have been serving in South Korea as **Professional Research Personnel (Alternative Military Service)**, with an expected completion in 2028.

I am highly motivated to analyze **financial phenomena through data-driven research**. Throughout my undergraduate and graduate curriculum, I have utilized a wide range of approaches from classical statistical techniques to state-of-the-art AI. With these experiences, I'm eager to study financial phenomena from two complementary angles: **(i) an engineering viewpoint** focused on improving analytical performance and **(ii) an economic perspective** aimed at deriving practical, real-world implications. With this background and these research interests, I aim to make meaningful contributions to the field.

* This CV is updated on 23, Aug, 2025.

RESEARCH INTERESTS

Financial Domain; Data Science; Statistical Modeling; Generative AI; Deep Learning.

EDUCATION

• POSTECH <i>M.S. in Dept. of Industrial and Management Engineering</i> - Industrial AI program Thesis Title: Systemic Cyber Risks and Insurance Regulatory Capital.	Pohang, South Korea <i>Sep. 2023 - Aug. 2025</i>
• Inha University <i>B.S. in Dept. of Industrial Engineering</i>	Incheon, South Korea <i>Mar. 2019 - Aug. 2023</i>

EXPERIENCE

• Sixty Hertz <i>Data Scientist (as a Technical Research Personnel)</i> - Performing anomaly detection using predictions of renewable energy yield from a VPP(Virtual Power Plant). Skill used: Python, SQL. Communication Tool: Slack / Jira, Confluence.	Seoul, South Korea <i>Jul. 2025 - <u>Current</u></i>
• AIRM Lab., POSTECH <i>Graduate Researcher (Advisor: Kwangmin Jung)</i> - Conducted scenario-based analyses on how systemic cyber risk affects insurance regulatory models. Teaching Assistant (TA): Financial Accounting (IMEN203), Spring 2024; Spring 2025. Skill used: Python, R. Communication Tool: Slack.	Pohang, South Korea <i>Sep. 2023 - Jul. 2025</i>
• Informatics and Deep Learning Lab., Inha Univ. <i>Research Intern (Advisor: Wookey Lee)</i> - Analyzed performance degradation in Deep Learning (e.g., Multi-modal and Diffusion generative models) under adversarial attacks on input distributions. Skill used: Python / Keras, TensorFlow.	Incheon, South Korea <i>Apr. 2021 - Jun. 2023</i>

• **SL Solution Co. Ltd.**

Seoul, South Korea

Internship course at Software Team

Aug. 2020

- Developed a **responsive web application** visualizing COVID-19 patient movement paths on interactive maps by integrating public datasets with **geocoding** through the **Tmap REST API**.
- **Skill used:** Java, Javascript / SpringBoot.

• **UDMTEK Co. Ltd.**

Suwon, South Korea

Internship course at MES Team

Dec. 2019 – Feb. 2020

- Managed database on **MES**(Manufacturing Execution System) and developed a **MVC**(Model, View, and Controller) service for maintenance.
- **Skill used:** Java, TypeScript / Angular, SpringBoot.

PUBLICATIONS

A. Working Papers

- [1] **Bae, K.**, Jung, K., and Zhang, L. (2025). Systemic cyber risks and insurance regulatory capital. Working Paper.

B. International Conferences

- [1] **Bae, K.**, Jung, K., and Zhang, L. (2025, Aug.). Systemic Cyber Risks and Insurance Regulatory Capital. *In the 5th World Risk and Insurance Economics Congress (**WRIEC**)*.
- [2] **Bae, K.**, Lee, S., and Lee, W. (2023, Dec.). Diffusion-C: Unveiling the Generative Challenges of Diffusion Models through Corrupted Data. *In the 37th Annual Conference on Neural Information Processing Systems (**NeurIPS**), Workshop on Diffusion Models*.
- [3] Lee, W., **Bae, K.**, Toshpulatov, M., Lee, S. (2022, Jul.). Bimodal Classification with CNN and Sequence Models. *In the 18th Int. Conference on Data Science (**ICDATA**)*.
- [4] **Bae, K.**, Lee, S., and Lee, W. (2022, Jan.). Transformer Networks for Trajectory Classification. *In the 2022 IEEE International Conference on Big Data and Smart Computing (**BigComp**)*.

C. Domestic Journals

- [1] Jung, K. and **Bae, K.** (2025). A Classification and Statistical Analysis on Systemic Cyber Risks. *Korean Journal of Insurance (**KJI**)*, 142, p.1-34.

D. Domestic Conferences

- [1] **Bae, K.**, Jung, K., Zhang, L. (2025, Aug.). Systemic cyber risks and insurance regulatory capital. *In the Korean Insurance Academic Society (**KIAS**)*.
- [2] Jung, K. and **Bae, K.** (2025, Feb.). A Classification and Statistical Analysis on Systemic Cyber Risks. *In the 3rd Social Science Korea (**SSK**) Networking Symposium*.
- [3] Jung, K. and **Bae, K.** (2025, Feb.). A Classification and Statistical Analysis on Systemic Cyber Risks. *In the Korean Insurance Academic Society (**KIAS**)*.
- [4] **Bae, K.**, Lee, S., and Lee, W. (2022, Dec.). Robustness Analysis of Diffusion Generation model Using Noises and Corruptions. *In the Korean Software Congress (**KSC**)*.
- [5] **Bae, K.**, Lee, S., and Lee, W. (2022, Jul.). Robust Multimodal Classification Model Using Homogeneous Features. *In the Korean Computer Congress (**KCC**)*.

PROJECTS

- **Assessment of the Corporate Customer Service and Proposal for the Improvement.**

Korean Fire Protection Association (KFPA)

Sep. 2024 - Nov. 2024

- **Keywords:** Optimization, Linear Programming.
- Analyzed the attributes of KFPA inspection staff based on employment types (Subsidiary, Contract worker, Outsourcing) and proposed workforce management strategies for enhancing inspection efficiency.
- Optimized the number of personnel for building inspections using Linear Programming.

- **Data-driven Evaluation for Safety Assessment of the KFPA and its Future Strategy.**

Korean Fire Protection Association (KFPA)

Apr. 2024 - Jul. 2024

- **Keywords:** Indexing, Random Forest, XAI.
- Conducted data-driven analysis to estimate the efficiency of digital transformation strategies in building inspections.
- Utilized a machine learning algorithm, such as random forest, to examine the impact of building features on inspection time and employed eXplainable AI (XAI) techniques for ensuring the interpretability of the results.

- **Responsive Web Application Development.**

SL Solution

Aug. 2020

- **Keywords:** REST API, Geocoding, Responsive web service.
- Developed a responsive web application to store and visualize the location data of COVID-19 confirmed cases. [\[Github\]](#)
- Used the web development tools, including Spring Boot, MariaDB, and T-Map API.

- **MES Web Application Maintenance Project.**

UDMTEK

Dec. 2019 - Feb. 2020

- **Keywords:** MES service, Responsive web service.
- Developed a web application for Manufacturing Execution System (MES) services.
- Implemented the MVC pattern using web development frameworks (e.g., Angular, Spring Boot, and PostgreSQL).

SKILLS

- **Language:** Korean (Native), English (Advanced).
- **Programming:** Python, R, C++ (Advanced), Java, JavaScript, SQL (Intermediate).
- **AI Framework:** TensorFlow, PyTorch, Scikit-Learn (Advanced).